

Computational Fluid Dynamics - 7/2/2022

Prof. Parolini and Prof. Valdettaro

Exercise 1. We want to simulate the flow around an airfoil with an incidence angle $\beta = 5$ degrees, as represented in Figure 1. The kinematic viscosity of the fluid is $\nu = 0.01$. The problem is governed by the time-dependent incompressible Navier-Stokes equations. We consider the following boundary conditions:

- a uniform horizontal velocity profile is imposed on the inflow boundary Γ_4 : $\mathbf{u} = (1, 0)$;
- a homogeneous Neumann condition is imposed on the outflow boundary Γ_2 : $\nu \frac{\partial \mathbf{u}}{\partial \mathbf{n}} - p \mathbf{n} = \mathbf{0}$;
- free-slip velocity conditions (equivalent to symmetry conditions) are imposed on the lower and upper boundaries Γ_1 and Γ_3 : $u_y = 0$, $\frac{\partial u_x}{\partial \mathbf{n}} = 0$;
- a no-slip condition is imposed on the airfoil wall Γ_5 : $\mathbf{u} = (0, 0)$;

The problem is initialized considering, at time $t = 0$, the Stokes solution obtained prescribing the same boundary conditions as above.

The problem should be solved using the incremental Chorin-Temam projection method using P_2/P_1 finite elements for velocity and pressure and advancing in time until the instant $T_f = 40$ with a time step $\Delta t = 1$.

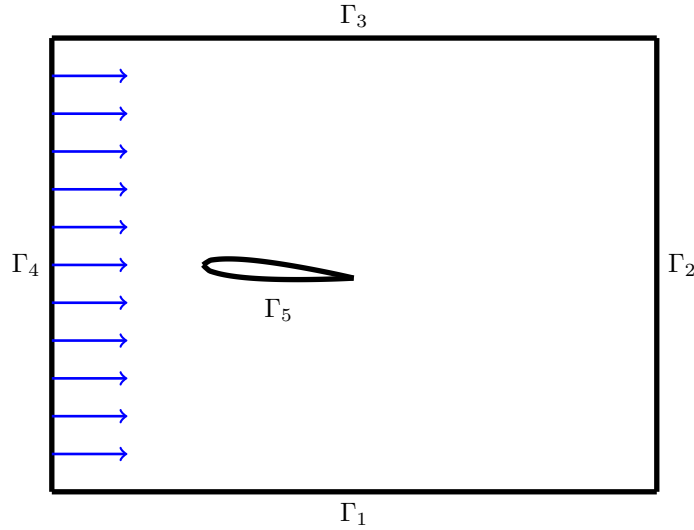


Figure 1: Computational domain

- Introduce the incremental Chorin-Temam projection method for the solution of the problem.
- Write the weak formulation of each sub-step of the projection method discussing how the boundary conditions associated to each sub-step are imposed.
- Based on the template `exam.edp`, write a `FreeFEM++` script for the solution of the problem.
- Estimate the force acting on the airfoil at time $T_f = 40$ for different values of the incidence angle $\beta = 0, 5, 10, 15, 20$ degrees. In particular, compute the drag force (the component in the direction x aligned with the freestream flow) and the lift force (the component in the direction y). (*Hint*: the forces can be computed integrating the stress on the wall boundary, whose normal vector components are defined in `FreeFEM++` as `N.x`, `N.y`).